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नेपाल इन्जिनियरिङ्ग सेवा, सिभिल समूह, विभिन्न (जनरल हाइवे , स्यानिटरी, इरिगेशन, हाईड्रोपावर) उपसमूह,
राजपत्राङ्कित तृतीय श्रेणी (प्राविधिक) पदको प्रतियोगितात्मक लिखित परीक्षा

मिति: २०८०/१०/२०

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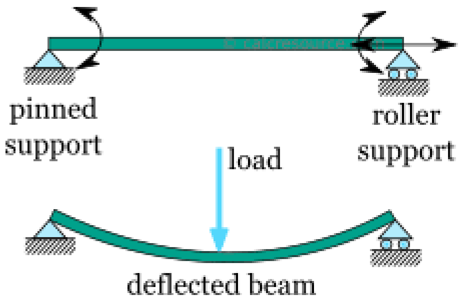
विषय: सेवा सम्बन्धी

Section-A (25 Marks)

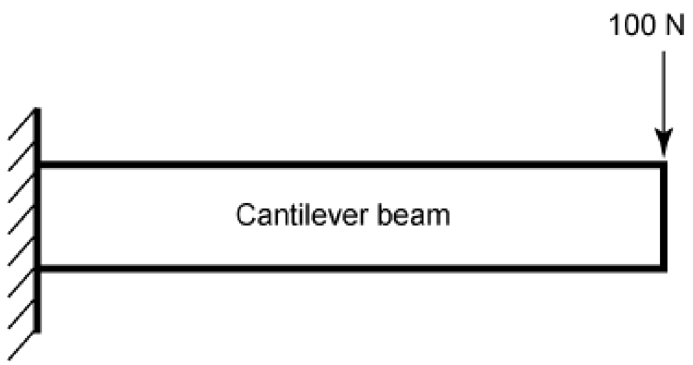
1. **Define simply supported supported and cantilever beam. Derive maximum bending moment for the beams in uniformly loaded condition. [2+8=10]**

Answer:

A simply supported beam is a beam that is supported at both ends, usually with one pin (hinge) support and one roller support. The supports prevent vertical movement but allow the beam to rotate freely at the ends. It does not resist bending moments at the supports.



A cantilever beam is a beam that is fixed at one end and free at the other end**.** The fixed end resists vertical forces, horizontal forces, and bending moments.



Simply Supported Beam Carrying a Uniformly Distributed Load (UDL)

Given,

Span of beam= L

Uniformly disturbed load= w KN/m over the entire span

Step 1: Calculate support reactions

Toral load on the beam=

$$W = wL$$

Since, the beam is symmetric,

$$R_A = R_B = \frac{wL}{2}$$

Step 2: Bending Moment at a section

Consider a section at a distance x from the left support.

The bending moment at that section is

$$M_x = R_A x - \frac{wx^2}{2}$$

$$Substituting\ R_A = \frac{wL}{2},$$

$$M_x = \frac{wLx}{2} - \frac{wx^2}{2}$$

Step 3: Determine Maximum Bending Moment

For maximum bending moment,

$$\frac{dM_x}{dx} = 0$$

$$\frac{d}{dx}\left(\frac{wLx}{2} - \frac{wx^2}{2}\right) = 0$$

$$\frac{wL}{2} - wx = 0$$

$$x = \frac{L}{2}$$

Thus, the maximum bending moment occurs at the centre of the beam.

Step 4: Substitute $x = \frac{L}{2}$

$$M_{max} = \frac{wL}{2}\left(\frac{L}{2}\right) - \frac{w}{2}\left(\frac{L}{2}\right)^2$$

$$= \frac{wL^2}{4} - \frac{wL^2}{8}$$

$$\boxed{M_{max} = \frac{wL^2}{8}}$$

Maximum Shear force: $V_{max} = \frac{wL}{2}$

Cantilever Beam carrying uniformly distributed Load (UDL)

Given:

Length of cantilever=L

Uniformly distributed load= w KN/m over the entire length

Step 1: Total load

$$W = wL$$

This acts at the centroid of the load i.e. $\frac{L}{2}$ from the fixed end.

Step 2: Fixed-end moment

Bending moment at the fixed support is:

$$M_{max} = W \times \frac{L}{2}$$

Substituting, $W = wL,$

$$M_{max} = wL\left(\frac{L}{2}\right)$$

$$\boxed{M_{max} = \frac{wL^2}{2}}$$

$$Negative\ sign\ indicates\ hogging$$

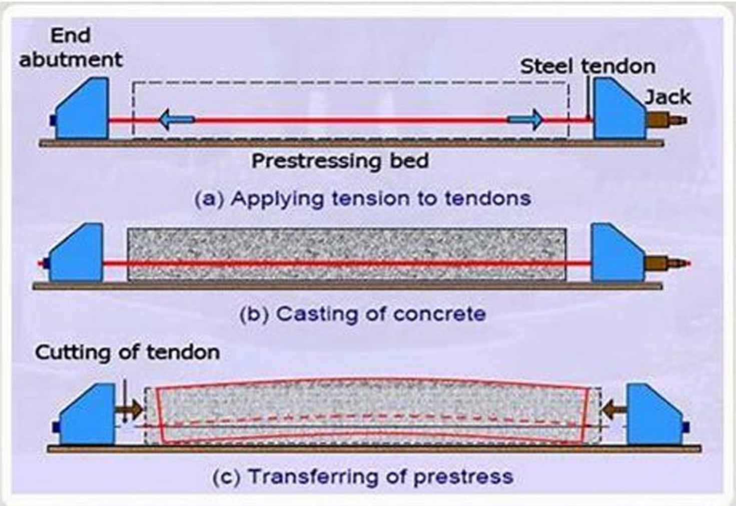
$$\boxed{M_{max} = -\frac{wL^2}{2}}$$

Maximum shear force: $V_{max} = wL$

2. **Describe the advantages and disadvantages of prestressed concrete. [10]**

Answer:

Prestressed concrete is the type of concrete in which internal compressive stresses are introduced by tensioning high-strength steel tendons before or after loading, so that tensile stresses caused by external loads are reduced or eliminated. Prestressed concrete is widely used for bridges, flyovers, long-span roofs, tanks, and other structures requiring high strength, durability, and longer spans, despite its higher initial cost and construction complexity.



Advantages of prestressed concrete:

i. Higher load carrying capacity:
Prestressed concrete can carry heavier loads than ordinary reinforced concrete (RCC).

ii. Economical for longer span structures:
Prestressed concrete is generally suitable for long-span bridges, roofs, flyovers and auditorium and is economical for long span structures.

iii. Reduction in cracking:
Prestressing keeps the concrete in compression which helps in minimizing tensile cracks and improves resistance to corrosion, weathering and water penetration.

iv. Results in smaller deflection in structure:
Deflections under service loads are significantly reduced in prestressed concrete.

v. Improved shear and fatigue resistance:
Prestressed concrete performs better under repeated and dynamic loading.

vi. Results in lighter structural members:
Prestressed concrete results in smaller depth beams which reduce the self-weight of the structure.

vii. Aesthetic appearance:
It results in slender sections which provides aesthetically pleasing structure.

Disadvantages of prestressed concrete:

i. High initial cost:
Prestressing equipment and high-strength materials are expensive.

ii. Requires skilled labor:
Prestressed concrete requires trained personnel for stressing and anchoring operations.

iii. Requires special equipment:
Hydraulic jacks, anchorages, ducts, and grouting equipment are necessary for prestressed concrete.

iv. Quality control:
Poor workmanship may lead to prestress losses or structural failure.

v. Prestress losses:
Losses in prestressed concrete occur due to creep, shrinkage, relaxation of steel, and friction, which must be accounted for in design.

vi. Uneconomical for short span:
For short-span structures, conventional RCC is generally more economical than prestressed concrete.

3. What forces should we consider while designing the retaining wall to stabilize the slope during road construction in hills? Enumerate. [10]

Answer:

A retaining wall is a structure designed to retain soil at different ground levels, resisting lateral earth pressure to provide stability for embankments, highways, and foundations. While designing a retaining wall in hilly road construction, all forces acting on the wall must be considered to ensure stability against sliding, overturning, bearing failure, and structural failure.

Several types of retaining walls are commonly used in the hill roads of Nepal. Some of the types of retaining walls constructed in Nepal are:

i. **Gabion Retaining Wall**
   a. Widely used along the Mugling–Narayanghat Highway and many rural hill roads.
   b. Flexible, economical, and performs well on weak foundations.
   c. Suitable where drainage is important.

ii. **Reinforced Concrete Cantilever Retaining Wall**
   a. Common on the Prithvi Highway and urban roads.
   b. Used for heights generally between 3–8 m.
   c. Durable but requires good foundation conditions.

iii. **Breast Wall**
   a. Constructed on the uphill side of roads to support cut slopes and prevent rockfalls.

b. Frequently seen on mountain highways.

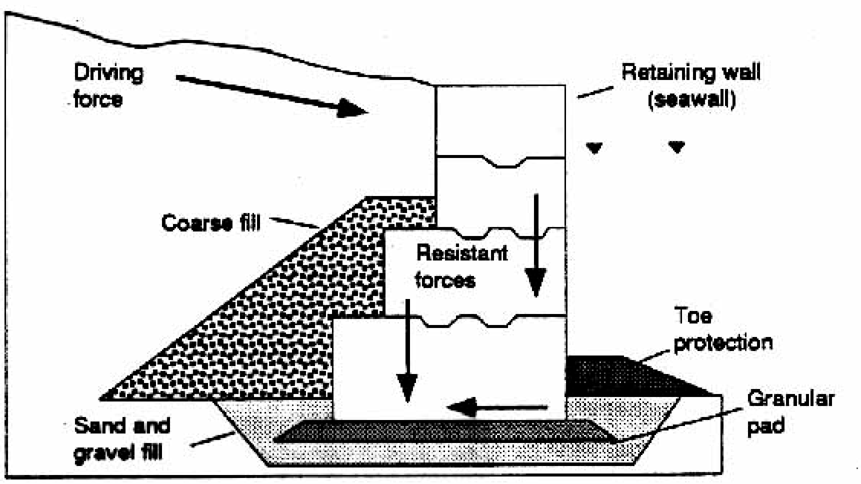
Along the Mugling–Narayanghat Highway, many unstable slopes are stabilized using gabion retaining walls combined with:

- Weep holes for drainage,
- Filter media behind the wall,
- Toe protection,
- Rock bolting and wire mesh on steep cut slopes.

This combination is effective because gabion walls are flexible, permeable, and better able to accommodate minor settlements and slope movements than rigid concrete walls.

The main forces that should be considered while designing the retaining wall during road construction in hills are described below in brief:

FIGURE 9: COMMON GRAVITY RETAINING WALLS



i. **Self-weight of the retaining wall (W)**
   a. Acts vertically downward through the center of gravity.
   b. Provides resistance against sliding and overturning.

ii. **Lateral earth pressure (Pa)**
   a. Exerted by the retained soil.
   b. Usually, the active earth pressure is considered (using Rankine's or Coulomb's theory).
   c. It is the primary force causing overturning and sliding.

iii. **Surcharge load (Ps)**
   a. Additional horizontal pressure due to loads on the backfill, such as:
      i. Traffic loads
      ii. Construction equipment
      iii. Buildings or structures
      iv. Stored materials
   b. Increases the lateral earth pressure.

iv. **Water pressure (Pw)**
   a. Hydrostatic pressure develops if water accumulates behind the wall due to poor drainage.
   b. Acts horizontally and significantly increases the overturning force.
   c. Proper drainage (weep holes, filter media, drainage pipes) is essential to reduce this pressure.

v. **Seismic (earthquake) forces (Pe)**
   a. Important in seismic regions like Nepal.
   b. Includes:
      i. Dynamic earth pressure on the backfill.

    ii. Inertial forces acting on the wall.
  c. Usually evaluated using the Mononobe–Okabe method.

vi. **Uplift pressure**
  a. Develops beneath the base due to groundwater.
  b. Reduces the effective weight of the wall and decreases stability.

vii. **Wind load (if applicable)**
  a. Considered for high retaining walls or walls with attached barriers or sound walls.
  b. Generally negligible for ordinary retaining walls.

viii. **Foundation reaction (Bearing pressure)**
  a. Upward reaction provided by the foundation soil.
  b. Must remain within the allowable bearing capacity and should not cause excessive settlement.

ix. **Frictional resistance at the base**
  a. Acts along the base of the wall.
  b. Resists sliding.
  c. Depends on the coefficient of friction between the foundation soil and the wall.

x. **Passive earth pressure (Pp)**
  a. Develops in front of the toe of the wall.
  b. Provides additional resistance against sliding.
  c. Often neglected or considered partially for conservative design.

xi. **Slope pressure (for sloping backfill)**
  a. Sloping ground behind the wall increases lateral earth pressure compared with level backfill.
  b. Must be accounted for in hill roads.

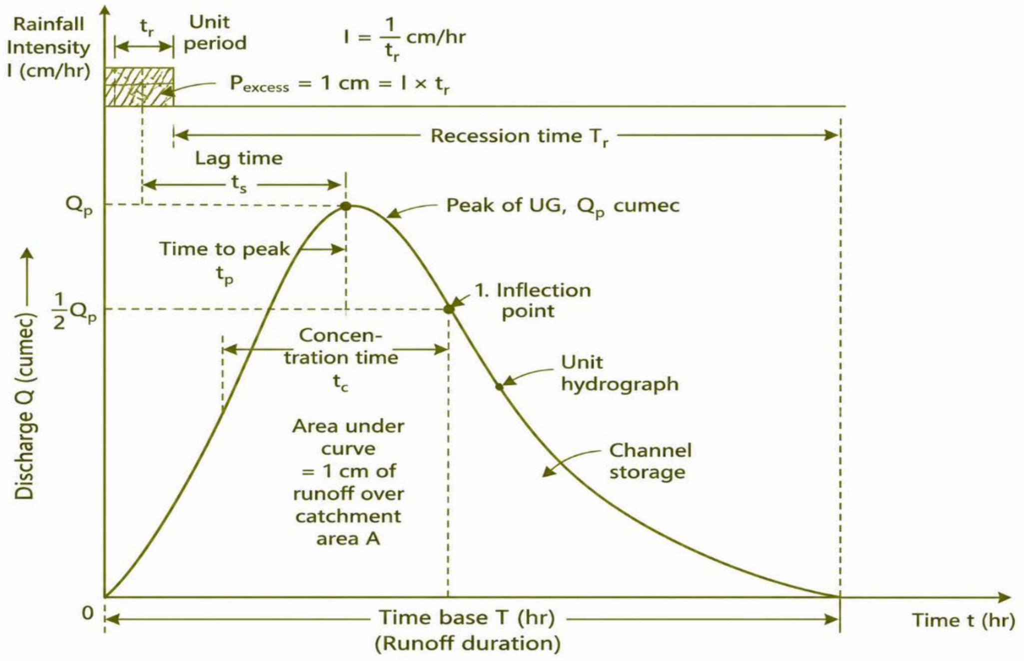
Section-B (25 Marks)

4. **What is a Unit Hydrograph (UH)? How do you estimate design flood using UH? Mention. [5]**

Answer:

Unit hydrograph is the direct run of hydrograph resulting from unit depth (1cm) of excess rainfall occurring at the uniform rate for the specific duration of D hours.

Unit hydrograph is the **unit pulse response function** of linear hydrologic system.



The basic assumptions of the unit hydrograph theory in flood forecasting in rivers of Nepal are listed below:

i. The excess rainfall has constant intensity one by $\frac{1}{D}$ cm/hr for effective storm duration of D hours.
ii. The excess rainfall is uniformly distributed over the entire catchment area.
iii. The base time of direct runoff hydrograph (DRH) resulting from excess rainfall of given duration is constant.

Steps for estimating design flood using UH is mentioned below:

i. **Determine the Design Storm:**
   - The design rainfall corresponding to the desired return period (e.g., 25, 50, or 100 years) is selected.
   - The rainfall depth and duration from Intensity-Duration-Frequency (IDF) data is obtained.

ii. **Estimate Effective Rainfall**
   - Different type of losses such as interception, infiltration, and depression storage from total rainfall is deducted.
   - Effective rainfall = Total rainfall − Losses.

iii. **Divide Effective Rainfall into Time Intervals**
   - The effective rainfall hyetograph is divided into increments corresponding to the duration of the unit hydrograph.

iv. **Apply Unit Hydrograph Ordinances**
   - Each rainfall excess increment is multiplied by the ordinates of the unit hydrograph.

v. **Perform Convolution (Superposition)**
   - The resulting hydrographs from each rainfall excess increment is added to obtain the Direct Runoff Hydrograph (DRH)**.**

vi. **Add Base Flow**
   - The estimated base flow on the DRH is superimposed.

vii. **Obtain the Design Flood Hydrograph**
   - The resulting hydrograph represents the design flood.
   - The maximum ordinate (peak discharge) of this hydrograph is taken as the design flood peak**.**

For a series of rainfall excesses:

$$Qn = P_1U_n + P_2U_{n-1} + P_3U_{n-2} + \cdots + P_mU_{n-m+1}Qn$$

$$Where:$$

$$Q_n = Direct\ runoff\ ordinate\ at\ time\ n$$

$$P_i = Effective\ rainfall\ depth\ during\ interval\ i$$

$$U_i = Unit\ hydrograph\ ordinate$$

$$m\ =\ Number\ of\ rainfall\ excess\ intervals$$

5. Assume that you are assigned with a responsibility of investigating a case of failure of a bridge immediately after its construction. What steps would you follow to investigate whether hydrology is a potential cause of the failure? Illustrate the steps with appropriate examples. Also, show a template of

report (with notes on expected contents under different chapters/sub-chapters) of such investigation to submit to a higher authority. [7+3=10]

Answer:

When a bridge fails immediately after construction, it is essential to determine whether hydrological factors such as flood discharge, scour, river morphology, or debris flow contributed to the failure. The investigation follows a systematic engineering procedure.

Steps for investigation:

Step1: Preliminary site inspection

The main objective of this stage of investigation is to assess the nature and extent of bridge failure.

Step 2: Collection of background information

All the available documents are gathered:

- Bridge design drawings
- Hydrological design report
- Geological investigation report and so on

Step 3: Review hydrological design

The original hydrological analysis must be verified.

Step 4: Flood and rainfall data analysis:

Collect:

- Rainfall records
- River gauge records
- Satellite rainfall data
- Meteorological information

Step 5: hydraulic investigation

River hydraulics such as flow velocity, water surface profile etc. are studied and analyzed.

Step 6: Scour investigation

Scour is one of the most common causes of bridge failure.

Investigate:

- General scour
- Local pier scour
- Contraction scour

Step 7: River morphology:

The river course changes are observed.

Investigate:

- River migration
- Channel shifting
- Bank erosion
- Sediment deposition

Step 8: Check debris and sediment effect:

Assess:

- Driftwood accumulation
- Boulder impact
- Sediment deposition
- Landslide material

Step 9: Numerical and hydraulic modeling

We can use software such as for numerical and hydraulic modeling:

i. HEC-RAS

ii. HEC-HMS
iii. MIKE-11
iv. SWAT

Step 10: Determination of cause of failure

Evaluation is done whether failure resulted from:

- Underestimated flood
- Inadequate waterway
- Insufficient freeboard
- Excessive scour

Suppose a newly constructed bridge over a hill river in Nepal collapsed during the first monsoon. Investigation found:

- Actual flood = 1300 m^3/s
- Design flood = 850 m^3/s
- Pier scour = 4.8 m
- Foundation depth = 4.0 m
- Large debris blocked bridge opening

The following conclusion can be drawn:

Failure was primarily caused by underestimated hydrological design combined with severe local scour and debris blockage.

Template of investigation report:

Chapter 1: Introduction

1.1: Background of project

1.2: Objective of investigation:

1.3: Scope of work

1.4: Seismicity of site

Chapter 2: Methodology

2.1: Boring

2.2: Sampling

2.3: Insitu test

2.4: Observation of GWT

2.5: Lab test

2.6: Analysis theory

i. Bearing capacity
ii. Slope stability
iii. Liquefaction potential

Chapter 3: Findings

3.1: Strata

3.2: GWT

3.3: Engineering properties

3.4: Classification of soil

Chapter 4: Calculation and analysis

i. Bearing capacity
ii. Slope stability
iii. Liquefaction potential

Chapter 5: Recommendation

Size, shape and depth of foundation with bearing capacity

Preventive measures for slope stability, liquefaction.

Annex:

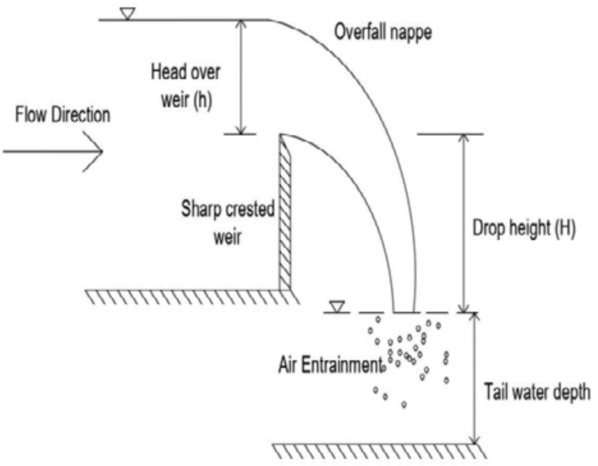
i. BH layout plan
ii. Borehole log
iii. Summary sheet of lab test
iv. Lab test for each sheet
v. Photographs of in-situ test, lab test etc.

6. What are the differences between weir and a barrage? What factors govern the design of a weir or a barrage? Explain. [4+6=10]

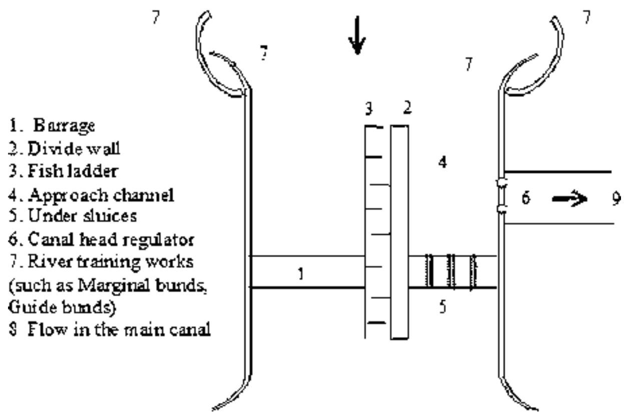
Answer:

A weir and a barrage are both hydraulic structures built across rivers to raise the upstream water level for irrigation, water supply, navigation, or flow regulation. However, they differ significantly in design, operation, and purpose. The main differences between weir and barrage is listed below:

Weir	Barrage
i. Weir is a fixed barrier constructed across a river to raise the water level.	i. Barrage is a gated structure constructed across a river with adjustable gates to regulate water flow.
ii. It is Usually a solid, permanent masonry or concrete structure.	ii. It consists of piers with movable steel gates between them.
iii. Water flows over the crest continuously; flow cannot be regulated.	iii. Flow is controlled by opening or closing the gates.
iv. Weir is fixed and non-adjustable.	iv. Barrage is Flexible and adjustable according to river conditions.
v. It has limited flood control capability.	v. It has effective flood regulation by operating gates during flood.
vi. Weir is less expensive due to its simpler operation.	vi. Barrage is more expensive because of the gate and hoisting equipment and control systems.



a. Weir



b. Barrage

The design of a weir or barrage depends on hydraulic, hydrological, geological, structural, and operational considerations. The objective is to ensure safe passage of floods, reliable water diversion, structural stability, and economical construction.

i. Hydrological factors:

Hydrological analysis determines the design discharge and flood characteristics.

- Catchment area and its characteristics
- Design flood (return period)
- Peak discharge

- Rainfall intensity and runoff
- Seasonal variation in river flow
- Climate change impacts (where applicable)

It helps to determine the size of the waterway, crest level, and flood-handling capacity.

ii. **Hydraulic factors:**
These ensure efficient flow and minimize energy losses.

- Water surface profile
- Flow velocity
- Afflux (rise in upstream water level)
- Head over the crest
- Energy dissipation arrangements
- Flow distribution through gates (for barrage)
- Sediment transport

It helps to prevent excessive erosion, overtopping, and inefficient flow conditions.

iii. **Geological and foundation conditions:**
The foundation must safely support the structure.

- Type and bearing capacity of foundation soil/rock
- Depth of hard strata
- Permeability of foundation
- Seepage characteristics
- Settlement potential
- Scour susceptibility

It helps to ensure stability against settlement, piping, and foundation failure.

iv. **Scour considerations:**
Flow around the structure causes erosion of the riverbed.

- Maximum expected scour depth
- Local scour near piers and abutments
- General scour
- Contraction scour

It helps to determine the depth of foundations and protective works.

v. **Economic considerations:**

- Construction cost
- Maintenance cost
- Availability of construction materials
- Construction duration
- Life-cycle cost

It ensures the project is economically feasible.

7. **Why is selecting the runway length important in airport? Mention. What factors influence the length of a runway in airport design? State. [5]**

Answer:

The runway length is one of the most critical design parameters in an airport which determines whether an aircraft can take off and land safely under all expected operating conditions**.** An inadequate runway length can compromise safety, restrict aircraft operations, and reduce the airport's operational efficiency. Selecting the runway length in airport is important because it:

i. **Ensures Safe Take-off**:
Runway length provides sufficient distance for an aircraft to accelerate and become airborne safely.

ii. **Ensures Safe Landing**:
It allows adequate distance for touchdown, braking, and stopping under normal and emergency conditions.

iii. **Accommodates Different Aircraft Types**:
The runway must be suitable for the largest and heaviest aircraft expected to use the airport.

iv. **Improves Flight Safety**:
It also reduces the risk of runway overruns, aborted take-offs, and landing accidents.

v. **Accounts for Environmental Conditions**

i. Runway length is adjusted for:
ii. Airport elevation (altitude)
iii. Air temperature
iv. Runway gradient (slope)
v. Wind conditions

The factors that influence the length of a runway in airport design are as follows:

i. Aircraft characteristics:

- Length of runway in airport design is influenced by:

  - Type and size of aircraft (design aircraft)
  - Maximum take-off weight (MTOW)
  - Landing weight
  - Engine performance
  - Aircraft speed during take-off and landing

- For example: Larger and heavier aircraft require longer runways.

ii. Airport elevation (altitude):

- Air density decreases with increasing altitude.
- Lower air density reduces engine thrust and aerodynamic lift.
- Airports at higher elevations require longer runways.
- The **ICAO recommendation** for correcting runway length is:

$$\boxed{L_e = L_0\left(1 + 0.07\frac{H}{300}\right)}$$

where:

$$L_e = Corrected\ runway\ length\ for\ elevation(m)$$

$$L_0 = Basic\ runway\ \ length\ at\ sea\ level(m)$$

$$H = Airport\ elevation\ above\ mean\ sea\ level(m)$$

- Increase the basic runway length by **7% for every 300 m rise in airport elevation**.

iii. Air temperature:

- High temperatures reduce air density (high density altitude).
- Aircraft need a higher take-off speed.
- Higher temperatures increase the required runway length.

$$The\ corrected\ runway\ length\ is:$$

$$L_T = L_E[1 + 0.01(T_r - T_s)]$$

$$Where:$$

$$L_T = runway\ length\ corrected\ for\ temperature(m)$$

$$L_E = runway\ length\ corrected\ for\ elevation(m)$$

$$T_r = airport\ reference\ temperature(°C)$$

$$T_s = standard\ atmospheric\ temperature\ at\ the\ airport\ elevation(°C)$$

- Increase the runway length by **1% for every 1°$\boldsymbol{C}$** that the airport reference temperature exceeds the standard atmospheric temperature.

iv. Runway gradient/slope:

- An uphill runway increases resistance during take-off.
- A downhill runway increases landing distance.
- Steeper runway gradients generally require longer runways.

$$Lg = L(1 + 0.20G)$$
$$Where:$$

$$L_g = runway\ length\ corrected\ for\ gradient(m)$$

$$L = runway\ length\ after\ elevation\ and\ temperature\ corrections(m)$$

$$G = effective\ runway\ gradient(\%)$$

- Increase runway length by **20% of the corrected runway length for every 1% effective runway gradient**.

v. Wind conditions:

- **Headwind** reduces the ground distance required for take-off and landing.
- **Tailwind** increases the required distance.
- Strong headwinds reduce runway length requirements, while tailwinds increase them.

Section-C (25 Marks)

8. Describe the factors to be considered in design of pavements. Explain about CBR method of flexible pavement design. [6+4=10]

Answer:

The main objective of pavement design is to provide a durable, safe, and economical pavement capable of carrying the anticipated traffic loads throughout its design life.

There are various design factors that are explained in brief:

a. Traffic loading:

- Higher traffic loads require thicker and stronger pavement layers.
- Traffic loading depends on:

  Magnitude and frequency of wheel loads.
  Number of commercial vehicles
  Axle load configuration (single, tandem, tridem).
  Traffic growth rate.
  Cumulative standard axles (CSA) or Equivalent Single Axle Loads (ESAL).

b. Subgrade soil characteristics:

- Weak subgrade requires improved soil or thicker pavement.
- It depends on:

  Soil type and classification.
  Bearing capacity (CBR for flexible pavement).
  Modulus of subgrade reaction (k-value) for rigid pavement.
  Compressibility and settlement characteristics.
  Swelling and shrinkage potential.

c. **Climate and environmental conditions:**

- ➢ Climate influences pavement strength, cracking, rutting, and durability.
- ➢ It depends on:

  Rainfall and drainage conditions.
  Temperature variation.
  Freeze–thaw cycles (cold regions).
  Moisture fluctuation.
  Solar radiation and humidity.

d. **Drainage conditions:**

- ➢ Poor drainage weakens the pavement and shortens its service life.
- ➢ It depends on:

  Surface drainage efficiency.
  Subsurface drainage.
  Groundwater level.
  Water infiltration through pavement joints and cracks.

e. **Pavement materials:**

- ➢ High-quality materials improve pavement performance and longevity.
- ➢ It depends on:

  Quality of aggregates.
  Bitumen or cement properties.
  Strength, stiffness, and durability.
  Availability of local materials.
  Resistance to weathering and wear.

f. **Design life:**

- ➢ Longer design life generally requires a stronger pavement structure.
- ➢ Design life is the expected service period before major rehabilitation.

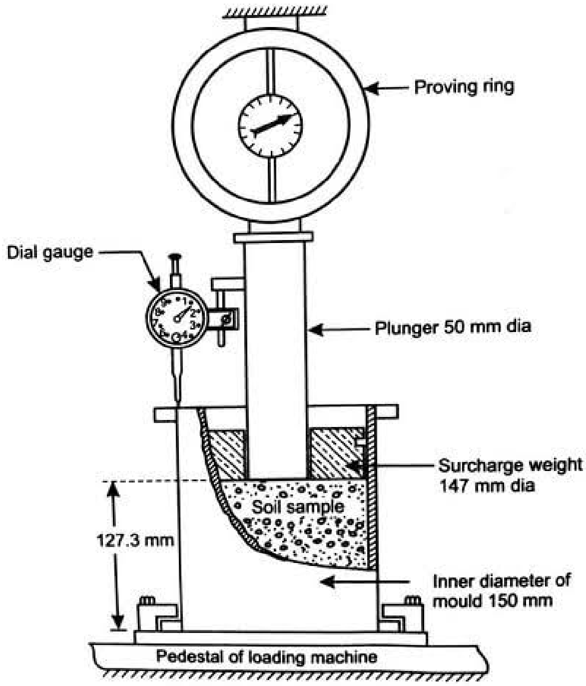
Common values:

  Flexible pavement: **15–20 years**
  Rigid pavement: **20–30 years**

California Bearing Ratio (CBR) method is an empirical method used for the design of flexible pavements**.** It determines the required pavement thickness based on the strength of the subgrade soil, expressed as the CBR value, and the expected traffic loading.

Principle:

- ➢ The method assumes that stronger subgrade soils (higher CBR) require thinner pavements, while weaker soils (lower CBR) require thicker pavements.
- ➢ The pavement thickness is selected from standard design charts or tables using the subgrade CBR and design traffic**.**



Steps in CBR method of pavement design:

i. **Conduct Traffic Survey**
   a. The expected traffic volume is estimated.
   b. The traffic is converted into cumulative standard axle loads (ESAL or msa).

ii. **Determine Subgrade Strength**
   a. The representative soil samples are collected.
   b. **CBR test** (laboratory soaked/unsoaked or field CBR test) is performed.
   c. The design CBR value (usually the soaked CBR for highways) is adopted.

iii. **Select Design Parameters**
   a. Suitable design life (e.g., 10–20 years) is selected.
   b. Traffic growth rate is estimated or determined.
   c. Environmental and drainage conditions are also considered.

iv. **Determine Total Pavement Thickness**
   a. The standard CBR design charts (e.g., IRC or FAA/other agency charts) corresponding to the design traffic and CBR value. Is used.
   b. The required total pavement thickness above the subgrade is read.

v. **Design Individual Pavement Layers**
   a. The total thickness is divided into:
i. **Wearing (surface) course**
ii. **Base course**
iii. **Sub-base course**
   b. Thickness of each layer is selected according to design standards and material quality.

vi. **Check Drainage and Construction Requirements**
   a. Proper drainage is ensured.
   b. Construction materials must be verified if they meet specification requirements.

9. What are the design criteria for designing a rigid pavement in a highway construction? Elucidate. Specify the conditions where rigid pavement is applicable. [10]

Answer:

Rigid pavement also known as cement concrete pavement is designed to withstand wheel loads, environmental effects, and repeated traffic throughout its design life. The main design criteria are as follows:

i. Design traffic:

- The cumulative number of commercial vehicles or axle load repetitions during the design life is estimated.
- The axle load spectrum, traffic growth rate, and lane distribution are considered.

ii. **Design life:**
- The intended service life of the pavement (typically **20–30 years** for highways) is selected.

iii. **Subgrade strength:**
- The bearing capacity of the subgrade is determined using:
  a. Modulus of subgrade reaction (**k-value**)
  b. CBR (converted to k-value if required)
- A stronger subgrade reduces slab thickness.

iv. **Concrete strength:**
- The flexural strength (modulus of rupture) of concrete is used as the primary design parameter.
- Also, compressive strength, modulus of elasticity, and Poisson's ratio are considered.

v. **Slab thickness:**
- The required slab thickness is calculated so that tensile stresses remain within allowable limits under traffic loading.

vi. **Temperature and moisture conditions:**
- Accounts for:
  a. Temperature gradients causing slab warping and curling.
  b. Moisture variation leading to slab deformation.

Rigid pavement is preferred where high strength, durability, and low maintenance are required. It is suitable under the following conditions:

i. **Heavy Traffic Loads**
- Highways carrying a large volume of heavy trucks and buses.
- Industrial roads and freight corridors.

ii. **High Traffic Intensity**
- Roads with high traffic density and frequent vehicle movements.
- Expressways and major urban roads.

iii. **Airport Pavements**
- Runways, taxiways, and aprons subjected to heavy aircraft wheel loads.

iv. **Poor Subgrade Conditions**
- Areas with weak or compressible soils where load distribution is critical.
- Rigid pavements bridge weak spots more effectively than flexible pavements.

v. **Intersections and Bus Stops**
- Locations with frequent braking, acceleration, and standing vehicles.
- Resistant to rutting and shoving.

vi. **Toll and Parking Areas**
- Areas subjected to slow-moving or stationary heavy loads.

vii. **Urban Roads**
- Roads where frequent maintenance and traffic disruption should be minimized.

viii. **Areas with High Rainfall or Waterlogging**
- Less susceptible to damage from water compared with bituminous pavements.

ix. **Hot Climate Regions**

- Concrete pavements are not softened by high temperatures, unlike asphalt pavements.

x. **Long Design Life Requirement**

- Projects requiring a service life of **30–40 years or more** with relatively low maintenance.

Section-D (25 Marks)

10. Draw a schematic diagram of a gravity flow water supply scheme and explain the function of each part. [5]

Answer:

In a gravity water supply system in rural hilly area of Nepal, source is usually fast flowing rivers and springs which is relatively free from harmful suspended materials. Hence, treatment plant is not incorporated in water supply project in rural hilly areas. However, in a gravity water supply system in rural terai regio, treatment plant must be incorporated to remove the suspended and other harmful materials from the water drawn from the source.

A flowchart of typical gravity Water Supply System (RWSS) in rural area of Nepal is shown below:

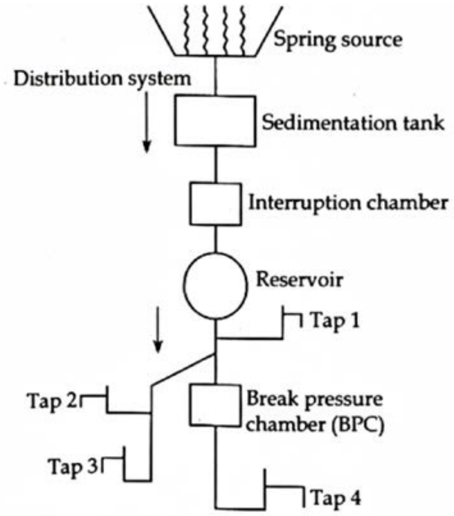


Figure: Typical rural water supply system

Components of water supply systems:

i. Source of water:
- Source of water should be reliable and contain minimum impurities.
- The amount of water available in the source must be adequate to tap the required amount to supply it to consumers.
- For rural hilly areas, source od water is springs which is considered to be relatively pure hence no treatment plant is provided in the water supply project od such areas.

ii. Intake:
- It is a device or structure constructed at or near the water source to draw the required amount of water to supply line.
- Must be easily accessible and must have minimum possibility of contamination.
- Primary function of intake is to draw the required amount of water from source.

iii. Distribution chamber:
- The main function of distribution chamber is to facilitate the divisions of flow in distribution subsystems.

iv. Transmission mains:
- The main function of transmission mains is to convey the water from source to treatment plant.

v. Interruption chamber:
- Interruption chamber is provided in transmission mains to break the pressure without float valve to prevent bursting of pipe due to excess pressure.

vi. Reservoir:
- The main function of reservoir is to store water from the source during low demand period and utilize it during high demand.

vii. Valve chamber:
- Chamber constructed to protect the valve in water supply projects.

viii. Distribution system:
- It is the pipe network to deliver water to consumer from the distribution reservoir.

ix. Break pressure tank:
- It is a pressure releasing device to break the pressure with float valve to prevent bursting of pipe due to excessive pressure.
- Provided in distribution mains.

11. Ideally, water supply system should run for 24 hours and 365 days a year. This system is known as continuous water supply system (CWS). But because of various reasons, many cities supply water few hours of a day or even per week. This system is called Intermittent water supply system (IWS). There is an agreement that CWS system is better than IWS in terms of network's durability and maintaining the water quality. Why the network's durability and water quality are compromised in IWS? Explain the reasons. [4+6=10]

Answer:

In an Intermittent Water Supply System (IWS)**,** water is supplied only for a limited number of hours each day or on certain days of the week. During non-supply periods, the distribution network is partially or completely depressurized. This repeated pressurization and depressurization adversely affect both the durability of the pipe network and the quality of the supplied water. In contrast, a Continuous Water Supply System (CWS) maintains positive pressure in the network at all times, reducing these problems.

Effect of IWS on Network Durability:

i. Pressure fluctuations (water hammer)
- Repeated filling and emptying of pipelines create sudden changes in pressure.
- These pressure surges (water hammer) impose additional stresses on pipes and joints.
- Water hammer phenomenon results in increased risk of pipe bursts, joint failures, and leakage.

ii. Frequent expansion and contraction:
- Cyclic pressurization causes pipes to expand and contract repeatedly.
- Frequent expansion and contraction cause fatigue of pipe materials, loosening of joints, and reduced service life.

iii. Increased leakage:
- Existing cracks and defective joints widen during pressure surges.
- Leakage becomes more severe with repeated pressure cycles.
- Increased leakage causes Greater water loss and higher maintenance costs.

iv. Sediment deposition and scouring:
- During stagnant periods, sediments settle inside the pipes.
- When supply resumes, higher flow velocities scour and transport these deposits.
- Sediment deposition causes internal abrasion, blockage, and increased wear of the distribution system.

v. Air entrapment:
- When the system is emptied, air enters the pipes.
- On resumption of supply, trapped air causes turbulence and additional pressure shocks.
- Air entrapment causes damage to pipelines, valves, and other appurtenances.

Effect of IWS on water quality:

i. **Contaminants in water:**
   - During supply interruptions, the pressure inside the pipes may become zero or negative.
   - Contaminated groundwater or sewage can enter the network through leaks and cracks.
   - Increased risk of contamination by bacteria, viruses, and other pathogens, leading to waterborne diseases.

ii. **Stagnation of water:**
   - Water remains stagnant in the pipelines during non-supply periods.
   - Stagnation of water causes:
     a. Loss of disinfectant residual (e.g., chlorine).
     b. Microbial regrowth and biofilm formation.
     c. Deterioration of taste and odor.

iii. **Corrosion:**
   - Alternate wetting and drying accelerate corrosion in metallic pipes.
   - Corrosion results in:
     a. Release of iron and other metals into the water.
     b. Further deterioration of water quality and reduction in pipe life.

iv. **Biofilm growth:**
   - Stagnant conditions encourage microorganisms to attach to the inner pipe surface and form biofilms.
   - It results in:
     a. Microbial contamination.
     b. Reduced effectiveness of disinfectants.
     c. Increased public health risk.

A continuous water supply system (CWS) is better than IWS as it:

- Maintains positive pressure throughout the network, preventing intrusion of contaminants.
- Reduces pressure fluctuations and water hammer.
- Minimizes sediment deposition and biofilm formation.
- Maintains adequate disinfectant residual.
- Improves the lifespan of pipes and fittings.
- Provides a reliable and safe water supply to consumers.

12. Point out the sources of water supply in Nepal. Describe the standard of drinking water quality prescribed by WHO. Also, clarify the processes for the treatment for treatment of water. [2+4+4=10]

Answer:

Nepal is the second richest country in terms of water resources. There are more than 6000 big and small rivers, numerous lakes, groundwater storage and glaciers that can prove to the major sources of water resources in Nepal. They are described as follows:

i. **Surface Water Sources**
   - Suitable for large urban and rural water supply schemes after proper treatment.
   - Rivers, streams, lakes and ponds, reservoirs, glacial snowmelt etc.

ii. **Groundwater Sources**
   - Springs (major source in hilly and mountainous regions), dug wells, tube wells, infiltration well and galleries etc.
   - Extensively used in the Terai and Kathmandu Valley.

iii. **Rainwater**
   - Rainwater harvesting from rooftops and catchment areas.
   - Commonly used in areas where conventional water sources are scarce, especially in remote hills and urban buildings.

World Health Organization (WHO) has prescribed guidelines for safe drinking water to protect public health. The water should be clear, colorless, pleasant in taste, free from harmful microorganisms, and free from toxic chemicals. The important WHO drinking water quality standards are summarized below:

a. Physical Quality Standards:

Parameter	WHO Guideline
Colour	≤ 15 TCU
Turbidity	≤ 5 NTU (preferably < 1 NTU)
Taste and Odour	Unobjectionable
Total Dissolved Solids (TDS)	≤ 1000 mg/L (desirable < 500 mg/L)

b. Chemical Quality standards:

Parameter	WHO Guideline Value
pH	6.5 – 8.5
Total Hardness	≤ 500 mg/L as $CaCO_3$
Chloride	≤ 250 mg/L
Sulphate	≤ 250 mg/L
Nitrate (NO_3^-)	≤ 50 mg/L
Fluoride	≤ 1.5 mg/L
Iron	≤ 0.3 mg/L
Manganese	≤ 0.1 mg/L
Arsenic	≤ 0.01 mg/L
Lead	≤ 0.01 mg/L
Cadmium	≤ 0.003 mg/L
Mercury	≤ 0.006 mg/L

c. Microbiological Quality Standards:

Parameter	WHO Guideline
E. coli / Thermotolerant coliform	0 per 100 mL
Total coliform bacteria	0 per 100 mL (for treated water)

Schematic flow diagram illustrating the possible treatment units involved in the water treatment process:

River Source
↓
Intake Structure
↓
Screening
↓
Plain Sedimentation
↓
Sedimentation with Coagulation
↓
Filtration
↓
Disinfection (Chlorination)
↓
Aeration
↓
Softening

↓
Miscellaneous

Intake structures:

- Hydraulic device constructed at or near river source to draw the required amount of water from the river source.
- Intake structures must be provided with the gate system to prevent entry of floating debris, large suspended sediments to the treatment plant.

Screening:

- Screening is the process of passing water through screens to remove large floating and suspended matter such as sticks, branches of tree, dead animals etc.
- Depending upon the size of the screens, there are three types of screens: coarse screen, fine screen and medium screen.
- Screening reduces the load on the treatment plant and increases the efficiency of treatment.

Plain sedimentation:

- The process of removal of coarse suspended particles present in the water such as sand, silt by allowing it to remain quiescent in sedimentation tank so that the suspended material settle down at the bottom by the action of gravity is known as plain sedimentation.
- Detention period in a plain sedimentation tank= 4-8 hours.
- SOR of a plain sedimentation tank=15 to 30 m^3/m^2/day

Sedimentation with coagulation:

- The fine suspended and colloidal materials like clay particles which cannot be removed by plain sedimentation or takes long time to removal by plain sedimentation can be removed by sedimentation with coagulation.
- Coagulants such as aluminum sulphate, sodium aluminate, iron salts are used.
- Detention period for sedimentation with coagulation = 1-4 hours
- SOR for sedimentation with coagulation= 30 to 40 m^3/m^2/day

Filtration:

- Filtration is the process of passing water through the beds of sand and granular material.
- For the removal of those impurities whose specific gravity is less than or equal to that of water for which sedimentation process does not work, in such case filtration is used.
- Slow sand filter, Rapid sand filter, pressure filters are some of the common filters used for the filtration of water in treatment plants.

Disinfection:

- The process of killing pathogenic bacteria and microorganisms present in water for the safe use is known as disinfection.
- The chemicals or substances used to kill microorganisms are disinfectants.
- Chlorination is one of the most common processes of disinfecting water.

Aeration:

- The process of bringing water in contact with air to remove dissolved gasses present in water is known as aeration.
- Aeration removes CO_2 gasses up to 70% due to which corrosion corrosiveness of water is reduced.

Softening:

- The process of removal or reduction of temporary or permanent hardness of water is known as softening.
- Softening helps to improve the taste of water, reduce consumption of soap during washing, bathing and also reduces formation of scales in boiler.

Miscellaneous:

- It includes:
  - Removal of iron and manganese
  - Removal of color, odor, taste from the water
  - Removal of arsenic and so on.

The treatment units are arranged in such a way that it progressively remove large debris, suspended solids, colloidal particles, microorganisms, and pathogens, to ensure that the water meets drinking water quality standards before distribution to consumers.